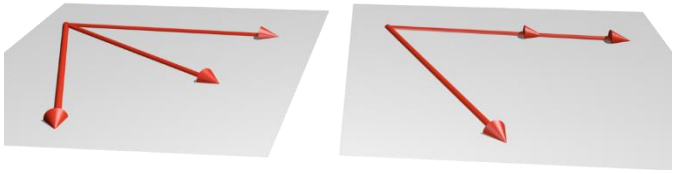
The Vector Equation of a Plane in R^3

Recall: If $\vec{w}$ is coplanar with $\vec{a}$ and $\vec{b}$, and $\vec{a}$ and $\vec{b}$ are not collinear, then we can express $\vec{w}$ as a _____ of $\vec{a}$ and $\vec{b}$.

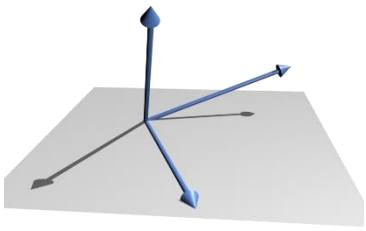
This is also called _____. A set of vectors is said to be linearly dependent if one of the vectors in the set can be defined as a linear combination of the others.

In R^2, if two non-zero vectors are _____, they will be considered as _____ and can form a _____ for the space.

In R^3, if three non-zero vectors are _____, they will be considered as _____ and can form a _____ for the space.



Linearly Dependent



Linearly Independent

Derive Vector Equation of a Plane in R^3

Let $\vec{a}$ and $\vec{b}$ be two _____ vectors (linearly independent) that lie in a plane.

Let $P_0(x_0, y_0, z_0)$ be a known point on the plane.

Let $P(x, y, z)$ be any point on the plane.

Vector Equation of a Plane in R^3

Parametric Equations of a Plane in R^3

Example 1: Determine the vector and parametric equations of the plane that contains the points $A(1,0,-3)$, $B(2,-3,1)$ and $C(3,5,-3)$.

Example 2: Determine the vector equation of a plane that contains the point $P(-4,3,6)$ and the line $L: \vec{r} = (1,-2,1) + t(2,-1,0)$, $t \in \Re$

Example 3: Does $A(4,5,-3)$ lie in the plane $\vec{r} = (4,1,6) + s(3,-2,1) + t(-6,6,-1)$, $s,t \in \Re$?

Challenge: Determine the point at which the plane $\vec{r} = (4,1,6) + s(3,-2,1) + t(-6,6,-1)$, $s,t \in \Re$ intersects the z-axis.